

Revant Nitin Bhatt

+91 9326076290 | rb3971@columbia.edu | 212 West 91st Street, New York, NY | LinkedIn: [Revant Nitin Bhatt](#) | revantnitinbhatt.com

EDUCATION

- Columbia University**, New York, NY Aug. 2026 - exp. Dec 2027
Master of Science in Mechanical Engineering, with a concentration in Robotics and Control
- Hochschule Rhein-Waal**, Kleve, DE Sept. 2021- Jun. 2025
Bachelor of Science in Mechatronics Systems Engineering, GPA: 3.52/4.0
- *Thesis: Characterization and Optimization of 3-D Printing Parameters for Polycaprolactone-based Medical Implants using a Magnetic Planar Drive-based 3-D Printer*
 - *Winner of the VDI University Prize for Outstanding Technical Thesis, 2026*

PROFESSIONAL EXPERIENCE

- Fraunhofer Institute of Production Technology (IPT)**, Bachelor Thesis Researcher, Aachen, DE Nov. 2024 – Jun. 2025
- Designed and executed a 19-run Box–Behnken experiment across 3 process parameters to optimize PCL scaffold printing on a Beckhoff XPlanar system.
 - Optimized and validated printing parameters through dimensional metrology, achieving approximately 97.1% mean dimensional accuracy and establishing a reproducible process protocol.
- Robert Bosch GmbH**, Corporate Research Intern, Renningen, DE Apr. 2024 – Sept. 2024
- Conducted fatigue and tensile tests on 84 samples in hydrogen and ambient environments to assess material durability and hydrogen embrittlement.
 - Developed MATLAB scripts to automate statistical post-processing and pattern detection across experimental datasets, streamlining repeatable analysis.
- Hochschule Rhein-Waal**, Student Teaching Assistant, DE Nov.2022 – Feb. 2024
- Delivered laboratory instruction and individual mentoring for MATLAB, Engineering Drawing and Design and Thermodynamics, resolving programming, SolidWorks and thermodynamics challenges.
- Hochschule Rhein-Waal**, Volunteer at Product Development Lab, Kleve, DE Apr. 2023 – Feb. 2024
- Maintained 6 FDM, SLA and SLS printers and prototyped electromechanical components for robotic-hand and bipedal-robot
- Danieli India Limited**, Engineering Intern, Chennai, IND Aug. 2022 – Sept. 2022
- Supported end-to-end production, assembly and QA/QC inspection of industrial cooling systems and nozzles.
- Lechler India Private Limited**, Manufacturing Intern, Mumbai, IND Feb. 2022 – Mar.2022
- Learned end-to-end production and assembly of cooling systems and nozzles; supported QA/QC inspections.

SELECTED PROJECTS

- VAJRA: RGB-Thermal Human-Robot Safety System** Sept. 2025 – Mar. 2026
- Developed an RGB–thermal vision-as-veto architecture for overhead ROI-based intrusion detection, achieving a mean stop-command latency of 35 ms; reanalyzing system logs and revising the manuscript for submission to an IEEE robotics conference.
- Tendon-Driven Robotic Hand** Mar. 2023 – Present
- Co-developed a tendon-driven robotic-hand prototype integrating real-time hand recognition using OpenCV and ASL gesture communication for assistive grasping and human–robot interaction.
- Finite Element Analysis of a Solar-Panel Hook** Sept. 2023 – Jan. 2024
- Analyzed static and cyclic loading in ANSYS to predict stress, deformation and fatigue performance and recommend reliability improvements

TECHNICAL SKILLS

Programming & Vision: MATLAB, Simulink, Python, C/C++, Microsoft Office, OpenCV
CAD & Simulation: SolidWorks, ANSYS Workbench, Fusion 360
Manufacturing & Testing: FDM/SLA/SLS, Design of Experiments, dimensional metrology, fatigue and tensile testing
Certifications: Certified SolidWorks Associate (CSWA)
Languages: English, German, Hindi, Gujarati, Marathi

PUBLICATIONS

Designing for Dynamics: Underactuation and Inertia as First Principles in Bipedal Robots, International Journal of Science and Research (IJSR), 14(11), 1285–1292 (Nov 2025). DOI: [10.21275/SR251117200057](https://doi.org/10.21275/SR251117200057)
Quantifying Serviceability: A Maintainability Framework for Tendon-Driven Robotic Hands, International Journal of Science and Research (IJSR), 14(9), 1527–1530 (Sep 2025). DOI: [10.21275/SR25929141237](https://doi.org/10.21275/SR25929141237)